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SPC 302 - Advanced Control Systems
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Abstract

Ride comfort and vehicle stability are fundamental objectives in automotive suspension design. Traditional passive suspension systems rely on fixed springs and dampers, which impose a compromise between passenger comfort and road holding performance. Active suspension systems address this limitation by introducing an external actuator capable of applying a controlled force to counteract road-induced vibrations.

In this project, a quarter-car active suspension model is developed to study the vertical dynamics of a vehicle. The equations of motion are derived using Newton's laws and expressed in state-space form. System properties such as controllability and observability are examined to verify the suitability of the model for control design. A state feedback controller is then designed to ensure that, under a 0.1 m step road disturbance, the car body oscillations remain within ± 0.1 m and decay within 5 seconds. Simulation results confirm that the designed controller successfully suppresses vibrations and improves ride comfort, demonstrating the effectiveness of active suspension control.

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Chapter 1

Introduction

Suspension systems are essential components in vehicles, serving to isolate the car body from road irregularities while maintaining continuous tire contact with the road surface. A well-functioning suspension system enhances passenger comfort, improves vehicle handling, and ensures driving safety. Conventional passive suspension systems employ fixed mechanical elements such as springs and dampers. While passive suspensions are simple and reliable, they cannot adapt to varying road conditions, resulting in a trade-off between ride comfort and stability.

Active suspension systems provide a modern solution by integrating sensors, controllers, and actuators to dynamically generate control forces that counteract road disturbances. This allows the vehicle body to remain relatively stable even when the wheels experience significant vertical motion due to uneven road surfaces. Consequently, active suspensions offer superior ride comfort and improved road handling compared to passive systems.

To analyze and design such systems efficiently, simplified mathematical models are often used. The quarter-car model represents one wheel of the vehicle along with the associated portion of the car body, capturing the essential vertical dynamics of the suspension system. In this project, the quarter-car active suspension model is studied in detail. The equations of motion are derived, transformed into state-space representation, and analyzed for controllability and observability. A state feedback controller is designed to meet specified ride comfort requirements, and the performance of the controlled system is validated through simulation. This study demonstrates how control theory can be effectively applied to enhance vehicle suspension performance and passenger comfort.

Chapter 2

Problem Description

Vehicle ride comfort and road handling are limited in passive suspension systems due to fixed spring and damper elements. Active suspension systems overcome this limitation by introducing a controllable actuator force that reduces vibrations caused by road disturbances.

In this project, a quarter-car active suspension model is studied. The system consists of a sprung mass (vehicle body), an unsprung mass (wheel assembly), suspension spring and damper, tire stiffness and damping, and an actuator applying a control force. The road displacement acts as an external disturbance input. The controlled output is the suspension deflection:

$$y_1 = x_1 - x_2,$$

which directly reflects passenger comfort.

The main objective is to model the system, derive its state-space representation, analyze its properties, and design state feedback controllers that satisfy the ride comfort requirement:

> *For a 0.1 m step road disturbance, the suspension deflection must remain within* ± 0.1 *m and settle within 5 seconds.*

Both pole-placement and LQR-based state feedback controllers are designed and validated using MATLAB and Simulink.

Chapter 3

Suspension System Description

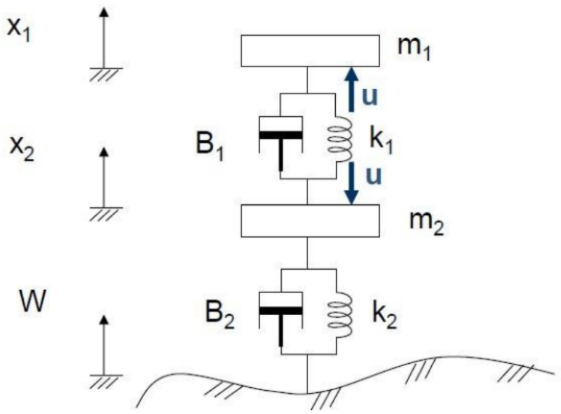


Figure 3.1: Quarter-car active suspension system model

Table 3.1: System Parameters

Parameter	**Units**
m_1 (Sprung mass)	kg
m_2 (Unsprung mass)	kg
k_1 (Suspension stiffness)	N/m
k_2 (Tire stiffness)	N/m
b_1 (Suspension damping)	N·s/m
b_2 (Tire damping)	N·s/m

3.1 Physical Meaning of Each Component in the Suspension Model

The model in Figure (1) represents a quarter-car active suspension system. This model describes one wheel and the portion of the car body supported by that wheel. Each element in the model corresponds directly to a real mechanical component found in an actual vehicle suspension system.

3.1.1 Sprung Mass M_1

The sprung mass represents the car body mass supported by the suspension, including the chassis, passengers, and cargo. Its vertical displacement x_1 describes how the car body moves up and down in response to road irregularities. Passenger comfort is directly related to minimizing oscillations of this mass.

3.1.2 Unsprung Mass M_2

The unsprung mass represents the wheel assembly, including the tire, rim, brake, and part of the axle. Its displacement x_2 describes how the wheel moves when encountering road disturbances.

3.1.3 Suspension Spring K_1

This is the suspension coil spring placed between the car body and the wheel assembly. It supports the vehicle weight and allows relative motion between M_1 and M_2. A softer spring improves comfort but reduces road holding, while a stiffer spring improves handling but transmits more vibration.

3.1.4 Suspension Damper b_1

This represents the shock absorber. It dissipates energy and damps oscillations of the suspension spring. Without this damper, the car would continue bouncing after hitting a bump.

3.1.5 Tire Spring K_2

This represents the tire stiffness, modeling how the tire deforms when contacting the road. It ensures the wheel maintains contact with the road surface.

3.1.6 Tire Damper b_2

This represents tire damping, accounting for energy loss in tire deformation and rubber hysteresis.

3.1.7 Road Disturbance W

This represents the road profile input, such as a bump or pothole. It acts as an external disturbance exciting the system.

3.1.8 Actuator Force U

This is the active control force generated by a hydraulic or electromagnetic actuator placed between the car body and the wheel. It is the key element that distinguishes active suspension from passive suspension. By applying controlled force, it counteracts vibrations and improves ride comfort.

3.1.9 Suspension Deflection $y_1 = x_1 - x_2$

This is the relative displacement between the car body and wheel, representing suspension travel. It is commonly measured in real vehicles using position sensors and is a critical feedback signal for control.

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3.2 Practical Implementation of the Control System

In a real active suspension system, several components work together to achieve controlled ride comfort.

3.2.1 Sensors

Position sensors measure suspension deflection $y_1 = x_1 - x_2$. Accelerometers measure the car body acceleration $\ddot{x}_1$. Wheel velocity sensors may measure $\dot{x}_2$.

These sensors provide real-time information about the suspension state to the controller.

3.2.2 Controller (Electronic Control Unit - ECU)

A microcontroller or digital signal processor receives sensor data and computes the required control force U using a control law such as state feedback or Linear Quadratic Regulator (LQR).

3.2.3 Actuator

A hydraulic actuator or electromagnetic linear motor applies the commanded force U between the sprung and unsprung masses. This actuator can push or pull to counteract road disturbances.

3.2.4 Power Supply

Active suspension systems require an external energy source such as a battery or hydraulic pump to provide power to the actuator.

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3.3 How Ride Comfort is Achieved

When the wheel encounters a road bump, the following sequence occurs:

- The road disturbance W excites the wheel mass M_2.
- Sensors detect suspension deflection and car body motion.
- The controller calculates a corrective force U.
- The actuator applies this force to oppose car body motion.

- Oscillations of the car body M_1 are rapidly reduced.

As a result:

- The car body returns quickly to equilibrium.
- Passenger-perceived vibration is minimized.
- The wheel maintains good contact with the road surface.

This demonstrates how active suspension control effectively improves ride comfort and vehicle stability.

Chapter 4

System Modeling

4.1 Modeling Equations of the Quarter-Car Active Suspension System

Figure 3.1 represents a quarter-car active suspension model. The system has two inputs:

- Control actuator force: $u(t)$
- Road displacement disturbance: $W(t)$

The vertical displacements of the sprung and unsprung masses are denoted by $x_1(t)$ and $x_2(t)$, respectively.

4.1.1 System Parameters

Table 4.1 summarizes the parameters used in this project and their selected values.

Table 4.1: System Parameters and Selected Values

Parameter	**Meaning**	**Reasonable Range**	**Selected Value**	**Units**
m_1	Sprung mass (1/4 of vehicle body)	$250 - 350$	300	kg
m_2	Unsprung mass (wheel + axle)	$30 - 60$	45	kg
b_1	Suspension damping coefficient	$1000 - 3000$	2000	N·s/m
b_2	Tire damping coefficient	$100 - 500$	300	N·s/m
k_1	Suspension spring stiffness	$15000 - 30000$	22500	N/m
k_2	Tire stiffness	$150000 - 250000$	200000	N/m

4.1.2 Equations of Motion (Time Domain)

Applying Newton's second law to each mass gives the following equations of motion.

Equation (1): Sprung Mass m_1

The forces acting between m_1 and m_2 are the suspension damper and spring, and the actuator force $u(t)$:

$$m_1\ddot{x}_1 = -b_1(\dot{x}_1 - \dot{x}_2) - k_1(x_1 - x_2) + u. \tag{4.1}$$

Rearranging (4.1) to express u:

$$u = m_1\ddot{x}_1 + b_1(\dot{x}_1 - \dot{x}_2) + k_1(x_1 - x_2). \tag{4.2}$$

Substituting the selected values from Table 4.1:

$$u = 300\,\ddot{x}_1 + 2000(\dot{x}_1 - \dot{x}_2) + 22500(x_1 - x_2). \tag{4.3}$$

Equation (2): Unsprung Mass m_2

The unsprung mass is affected by the suspension elements, tire elements, road input W, and the actuator force:

$$m_2\ddot{x}_2 = b_1(\dot{x}_1 - \dot{x}_2) + k_1(x_1 - x_2) + b_2(\dot{W} - \dot{x}_2) + k_2(W - x_2) - u. \tag{4.4}$$

Rearranging (4.4) to express $-u$:

$$-u = m_2\ddot{x}_2 + b_1(\dot{x}_2 - \dot{x}_1) + b_2(\dot{x}_2 - \dot{W}) + k_1(x_2 - x_1) + k_2(x_2 - W). \tag{4.5}$$

Substituting the selected values from Table 4.1:

$$-u = 45\,\ddot{x}_2 + 2000(\dot{x}_2 - \dot{x}_1) + 300(\dot{x}_2 - \dot{W}) + 22500(x_2 - x_1) + 200000(x_2 - W). \tag{4.6}$$

4.1.3 Laplace Transform (s-Domain Form)

Taking the Laplace transform of the time-domain equations (zero initial conditions) gives:

Equation (1) in the s-domain

Starting from (4.2):

$$\begin{aligned} U &= m_1s^2X_1 + b_1s(X_1 - X_2) + k_1(X_1 - X_2) \\ &= m_1s^2X_1 + b_1sX_1 - b_1sX_2 + k_1X_1 - k_1X_2 \\ &= (m_1s^2 + b_1s + k_1)X_1 - (b_1s + k_1)X_2. \end{aligned} \tag{4.7}$$

Substituting selected values:

$$U = (300s^2 + 2000s + 22500)X_1 - (2000s + 22500)X_2. \tag{4.8}$$

Equation (2) in the s-domain

From (4.5):

$$\begin{aligned} -U &= m_2s^2X_2 + b_1s(X_2 - X_1) + b_2s(X_2 - W) + k_1(X_2 - X_1) + k_2(X_2 - W) \\ &= m_2s^2X_2 + b_1sX_2 - b_1sX_1 + b_2sX_2 - b_2sW + k_1X_2 - k_1X_1 + k_2X_2 - k_2W \\ &= -(b_1s + k_1)X_1 + \big(m_2s^2 + (b_1 + b_2)s + (k_1 + k_2)\big)X_2 - (b_2s + k_2)W. \end{aligned} \tag{4.9}$$

Substituting selected values:

$$\begin{aligned} -U &= -(2000s + 22500)X_1 + \big(45s^2 + (2000 + 300)s + (22500 + 200000)\big)X_2 - (300s + 200000)W \\ &= -(2000s + 22500)X_1 + (45s^2 + 2300s + 222500)X_2 - (300s + 200000)W. \end{aligned} \tag{4.10}$$

4.1.4 Solving for X_1 in Terms of U and W (as in the derivation)

From (4.7), solve for X_2:

$$X_2 = \frac{(m_1 s^2 + b_1 s + k_1)X_1 - U}{b_1 s + k_1}. \tag{4.11}$$

Substitute (4.11) into (4.9):

$$-U = -(b_1 s + k_1)X_1 + \big(m_2 s^2 + (b_1 + b_2)s + (k_1 + k_2)\big)\left(\frac{(m_1 s^2 + b_1 s + k_1)X_1 - U}{b_1 s + k_1}\right) - (b_2 s + k_2)W. \tag{4.12}$$

Rearrange to isolate X_1:

$$X_1 = \frac{\big(m_2 s^2 + b_2 s + k_2\big)U + \big(b_1 s + k_1\big)\big(b_2 s + k_2\big)W}{\big(m_2 s^2 + (b_1 + b_2)s + (k_1 + k_2)\big)\big(m_1 s^2 + b_1 s + k_1\big) - (b_1 s + k_1)^2}. \tag{4.13}$$

Substituting the selected values:

$$X_1 = \frac{\big(45 s^2 + 300 s + 200000\big)U + \big(2000 s + 22500\big)\big(300 s + 200000\big)W}{\big(45 s^2 + 2300 s + 222500\big)\big(300 s^2 + 2000 s + 22500\big) - \big(2000 s + 22500\big)^2}. \tag{4.14}$$

4.1.5 Solving for X_2 in Terms of U and W

From Equation (4.7), we express X_1 in terms of X_2 and U:

$$X_1 = \frac{U + (b_1 s + k_1)X_2}{m_1 s^2 + b_1 s + k_1}. \tag{4.15}$$

Substituting (4.15) into Equation (4.9):

$$-U = -(b_1 s + k_1)\left(\frac{U + (b_1 s + k_1)X_2}{m_1 s^2 + b_1 s + k_1}\right) + \big(m_2 s^2 + (b_1 + b_2)s + (k_1 + k_2)\big)X_2 - (b_2 s + k_2)W. \tag{4.16}$$

Rearranging to isolate X_2:

$$X_2 = \frac{-m_1 s^2 U + (m_1 s^2 + b_1 s + k_1)(b_2 s + k_2)W}{\big(m_2 s^2 + (b_1 + b_2)s + (k_1 + k_2)\big)(m_1 s^2 + b_1 s + k_1) - (b_1 s + k_1)^2}. \tag{4.17}$$

Substituting the selected values:

$$X_2 = \frac{-300 s^2 U + (300 s^2 + 2000 s + 22500)(300 s + 200000)W}{(45 s^2 + 2300 s + 222500)(300 s^2 + 2000 s + 22500) - (2000 s + 22500)^2}. \tag{4.18}$$

4.1.6 Verification of the Input–Output Transfer Matrix

From (4.13) and (4.17), both $X_1(s)$ and $X_2(s)$ share the same denominator. Define the common denominator

$$\Delta(s) = \big(m_1 s^2 + b_1 s + k_1\big)\Big(m_2 s^2 + (b_1 + b_2)s + (k_1 + k_2)\Big) - \big(b_1 s + k_1\big)^2. \tag{4.19}$$

Using (4.13), we can separate the contributions of U and W:

$$X_1(s) = \frac{\big(m_2 s^2 + b_2 s + k_2\big)}{\Delta(s)}\, U(s) + \frac{\big(b_1 s + k_1\big)\big(b_2 s + k_2\big)}{\Delta(s)}\, W(s). \tag{4.20}$$

Now expand the product in the W-numerator:

$$\big(b_1 s + k_1\big)\big(b_2 s + k_2\big) = b_1 b_2 s^2 + (b_1 k_2 + b_2 k_1)s + k_1 k_2. \tag{4.21}$$

Similarly, from (4.17),

$$X_2(s) = \frac{-m_1 s^2}{\Delta(s)}\, U(s) + \frac{\big(m_1 s^2 + b_1 s + k_1\big)\big(b_2 s + k_2\big)}{\Delta(s)}\, W(s). \tag{4.22}$$

Expanding the W-numerator of (4.22):

$$\big(m_1 s^2 + b_1 s + k_1\big)\big(b_2 s + k_2\big) = m_1 b_2 s^3 + (m_1 k_2 + b_1 b_2)s^2 + (b_1 k_2 + b_2 k_1)s + k_1 k_2. \tag{4.23}$$

Therefore, the input–output relation can be written in matrix form as

$$\begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \frac{1}{\Delta(s)} \begin{bmatrix} m_2 s^2 + b_2 s + k_2 & b_1 b_2 s^2 + (b_1 k_2 + b_2 k_1)s + k_1 k_2 \\ -m_1 s^2 & m_1 b_2 s^3 + (m_1 k_2 + b_1 b_2)s^2 + (b_1 k_2 + b_2 k_1)s + k_1 k_2 \end{bmatrix} \begin{bmatrix} U(s) \\ W(s) \end{bmatrix}, \tag{4.24}$$

which matches the transfer matrix form requested.

Chapter 5

Transfer Function Analysis and Time-Domain Performance

5.1 Transfer Functions

The suspension deflection output is defined as:

$$Y_1(s) = X_1(s) - X_2(s). \tag{5.1}$$

We define the transfer functions:

$$G_1(s) = \frac{X_1(s) - X_2(s)}{U(s)}, \qquad G_2(s) = \frac{X_1(s) - X_2(s)}{W(s)}. \tag{5.2}$$

5.1.1 Expressions for X_1/U and X_2/U

From the Laplace-domain equations of motion, solving for X_1 and X_2 in terms of the control input U yields:

$$\frac{X_1(s)}{U(s)} = \frac{m_2 s^2 + b_2 s + k_2}{\Delta(s)}, \tag{5.3}$$

$$\frac{X_2(s)}{U(s)} = \frac{-m_1 s^2}{\Delta(s)}, \tag{5.4}$$

where the common denominator is

$$\Delta(s) = \big(m_2 s^2 + (b_1 + b_2)s + (k_1 + k_2)\big)\big(m_1 s^2 + b_1 s + k_1\big) - (b_1 s + k_1)^2. \tag{5.5}$$

5.1.2 Derivation of $G_1(s) = (X_1 - X_2)/U$

$$G_1(s) = \frac{X_1}{U} - \frac{X_2}{U} \tag{5.6}$$

$$= \frac{m_2 s^2 + b_2 s + k_2}{\Delta(s)} - \frac{-m_1 s^2}{\Delta(s)} \tag{5.7}$$

$$= \frac{m_2 s^2 + b_2 s + k_2 + m_1 s^2}{\Delta(s)} \tag{5.8}$$

$$= \frac{(m_1 + m_2)s^2 + b_2 s + k_2}{\Delta(s)}. \tag{5.9}$$

Thus,

$$G_1(s) = \frac{(m_1 + m_2)s^2 + b_2 s + k_2}{\big(m_2 s^2 + (b_1 + b_2)s + (k_1 + k_2)\big)\big(m_1 s^2 + b_1 s + k_1\big) - (b_1 s + k_1)^2}. \tag{5.10}$$

5.1.3 Expressions for X_1/W and X_2/W

Similarly, solving for the road disturbance input W gives:

$$\frac{X_1(s)}{W(s)} = \frac{(b_1 s + k_1)(b_2 s + k_2)}{\Delta(s)}, \tag{5.11}$$

$$\frac{X_2(s)}{W(s)} = \frac{(m_1 s^2 + b_1 s + k_1)(b_2 s + k_2)}{\Delta(s)}. \tag{5.12}$$

5.1.4 Derivation of $G_2(s) = (X_1 - X_2)/W$

$$G_2(s) = \frac{X_1}{W} - \frac{X_2}{W} \tag{5.13}$$

$$= \frac{(b_1 s + k_1)(b_2 s + k_2)}{\Delta(s)} - \frac{(m_1 s^2 + b_1 s + k_1)(b_2 s + k_2)}{\Delta(s)} \tag{5.14}$$

$$= \frac{\big[(b_1 s + k_1) - (m_1 s^2 + b_1 s + k_1)\big](b_2 s + k_2)}{\Delta(s)} \tag{5.15}$$

$$= \frac{-m_1 s^2 (b_2 s + k_2)}{\Delta(s)}. \tag{5.16}$$

Therefore,

$$G_2(s) = \frac{-m_1 s^2 (b_2 s + k_2)}{\big(m_2 s^2 + (b_1 + b_2)s + (k_1 + k_2)\big)\big(m_1 s^2 + b_1 s + k_1\big) - (b_1 s + k_1)^2}. \tag{5.17}$$

5.1.5 Substitution of Selected Numerical Parameters

Using the selected parameter values:

$$m_1 = 300, \quad m_2 = 45, \quad b_1 = 2000, \quad b_2 = 300, \quad k_1 = 22500, \quad k_2 = 200000,$$

we obtain:

$$G_1(s) = \frac{345s^2 + 300s + 200000}{(45s^2 + 2300s + 222500)(300s^2 + 2000s + 22500) - (2000s + 22500)^2}. \quad (5.18)$$

$$G_2(s) = \frac{-(300s^2)(300s + 200000)}{(45s^2 + 2300s + 222500)(300s^2 + 2000s + 22500) - (2000s + 22500)^2}. \quad (5.19)$$

5.2 Step Response Results

5.2.1 Printed Performance Metrics

Figure 5.1 shows the printed numerical performance metrics obtained from MATLAB for the transfer functions $G_1(s) = (X_1 - X_2)/U$ with unity step and $G_2(s) = (X_1 - X_2)/W$ with a step 0.1.

```
===== G1 = (X1 - X2)/U =====
Settling Time     = 1.324 s
Max Overshoot     = 29.26 %

===== G2 = (X1 - X2)/W =====
Settling Time     = NaN s
Max Overshoot     = Inf %

Damped Frequencies (rad/s):
G1: 7.94  7.94
G1: 63.40  63.40
G2: 7.94  7.94
G2: 63.40  63.40
```

Figure 5.1: Printed numerical performance metrics from MATLAB for $G_1(s)$ and $G_2(s)$.

5.2.2 Step Response of $G_1(s) = (X_1 - X_2)/U$

Figure 5.2 shows the MATLAB step response of the transfer function $G_1(s)$, which represents the suspension deflection due to the actuator control force input U.

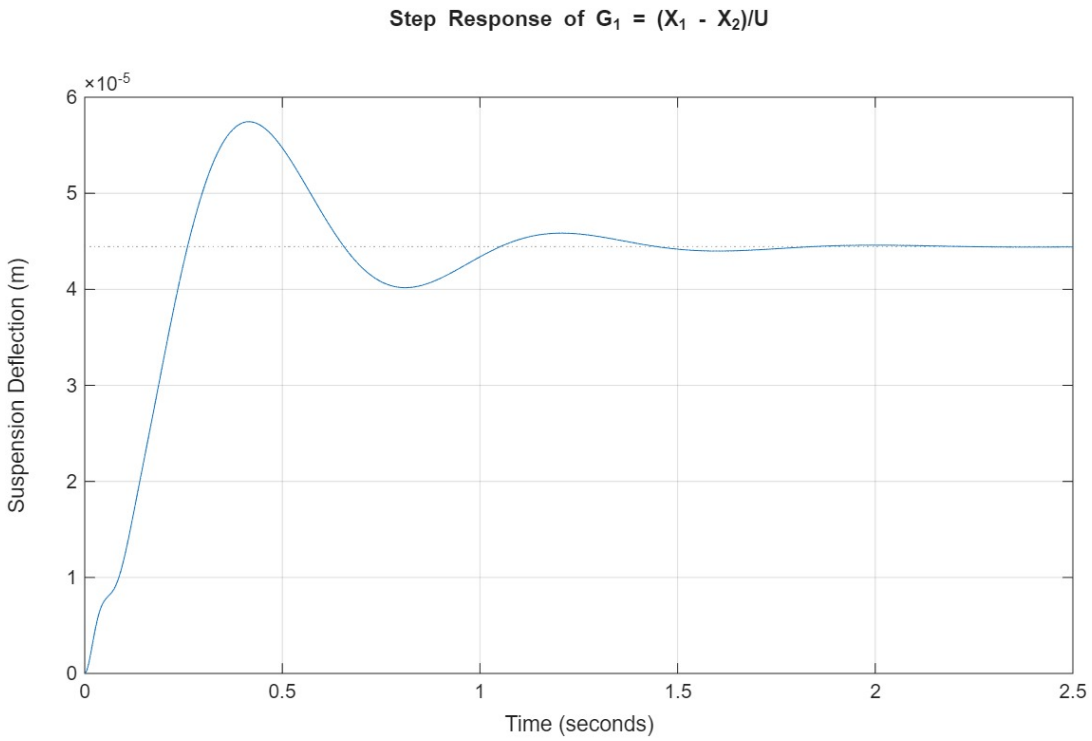


Figure 5.2: Step response of $G_1(s) = (X_1 - X_2)/U$ obtained from MATLAB.

5.2.3 Step Response of $G_2(s) = (X_1 - X_2)/W$ with $W = 0.1$ m

Figure 5.3 shows the MATLAB step response of the transfer function $G_2(s)$, which represents the suspension deflection due to the road disturbance input W. A step road displacement of $W = 0.1$ m was applied.

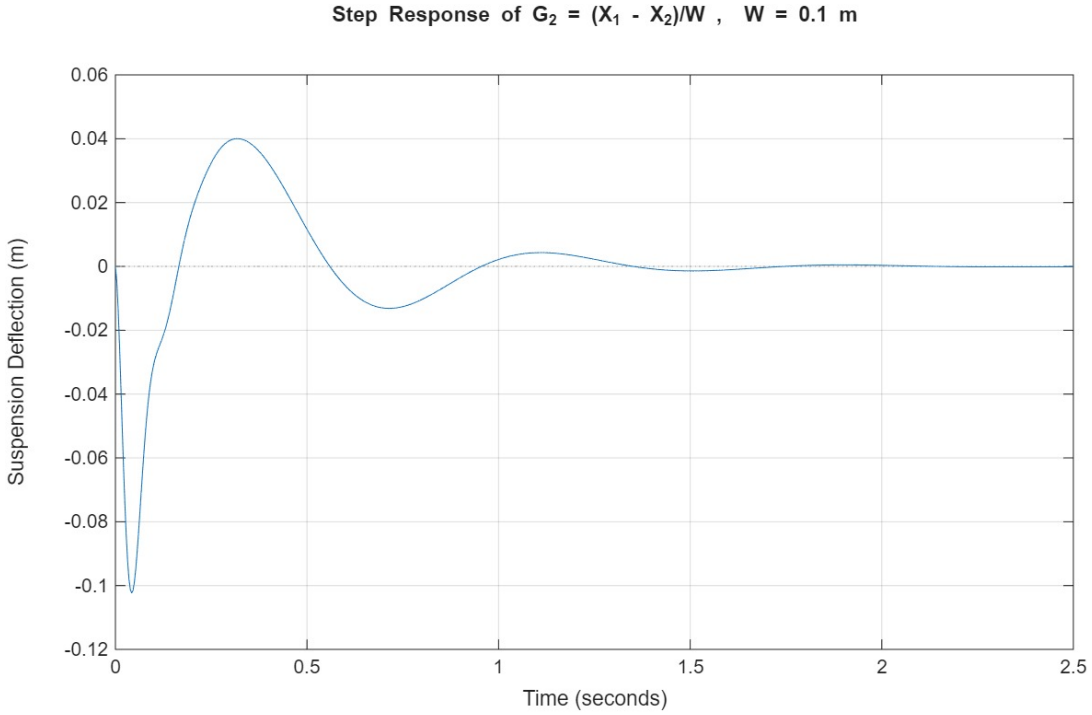


Figure 5.3: Step response of $G_2(s) = (X_1 - X_2)/W$ for a 0.1 m road step disturbance.

5.3 Discussion of Results

From the obtained MATLAB results:

- The figures above show the time responses of the suspension deflection for both transfer functions with respect to the actuator force input U and the road disturbance input W.
- For $G_1(s)$, the settling time is approximately 1.32 s and the maximum overshoot is about 29.3%, indicating a lightly damped response under actuator control input.
- For $G_2(s)$, the steady-state suspension deflection tends to zero after a step road disturbance, since both masses move together at steady state, resulting in zero relative displacement.
- Therefore, MATLAB may return NaN or infinite values for overshoot or settling time for $G_2(s)$ when using the default `stepinfo` definition around a zero final value.
- Both transfer functions share identical natural frequencies because they have the same denominator dynamics (same system poles).

Chapter 6

Open-Loop Stability Analysis

In this chapter, the stability of the open-loop quarter-car suspension system is investigated. Since both transfer functions $G_1(s)$ and $G_2(s)$ share the same denominator, the stability of the overall system is determined by examining the characteristic equation obtained from the common denominator.

6.1 Characteristic Equation

From the derived transfer functions, the common denominator is:

$$\Delta(s) = \big(m_2 s^2 + (b_1 + b_2)s + (k_1 + k_2)\big)\big(m_1 s^2 + b_1 s + k_1\big) - (b_1 s + k_1)^2. \tag{6.1}$$

Substituting the selected numerical parameters:

$$m_1 = 300, \quad m_2 = 45, \quad b_1 = 2000, \quad b_2 = 300, \quad k_1 = 22500, \quad k_2 = 200000,$$

the expanded characteristic polynomial becomes:

$$\Delta(s) = 1.35 \times 10^4 s^4 + 7.8 \times 10^5 s^3 + 6.83625 \times 10^6 s^2 + 4.0675 \times 10^7 s + 4.5 \times 10^9. \tag{6.2}$$

6.2 Open-Loop Poles

The poles of the open-loop system are the roots of $\Delta(s)$. Using MATLAB, the computed poles are:

$$s_{1,2} = -26.077 \pm 63.401j, \qquad s_{3,4} = -2.812 \pm 7.939j.$$

These poles are shown in the complex plane in the results image provided in the previous section.

6.3 Stability Verification

Is the Open-Loop System Stable?

Yes — the open-loop suspension system is stable.

All poles have negative real parts, which means they lie entirely in the left-half complex plane. Therefore, the system is asymptotically stable in open loop.

6.4 Physical Interpretation

The quarter-car model represents a passive mechanical mass–spring–damper system with:

- Positive stiffness coefficients (k_1, k_2),
- Positive damping coefficients (b_1, b_2),
- Positive masses (m_1, m_2).

Such systems naturally dissipate energy and cannot generate self-sustained oscillations. Hence, stability of the open-loop system is physically expected.

However, the complex conjugate poles with small negative real parts indicate a **lightly damped oscillatory response**, which justifies the need for active control to improve ride comfort and reduce oscillations.

6.5 Conclusion

The open-loop quarter-car suspension model is stable but lightly damped. Active feedback control will therefore be designed in the next chapter to achieve the required ride comfort specifications.

Chapter 7

State-Space Model

7.1 State-Space Representation with Output $y_1 = x_1 - x_2$

In this project, the suspension deflection is selected as the system output:

$$y_1 = x_1 - x_2. \tag{7.1}$$

Differentiating (7.1) gives:

$$\dot{y}_1 = \dot{x}_1 - \dot{x}_2 \quad \Rightarrow \quad \dot{x}_2 = \dot{x}_1 - \dot{y}_1. \tag{7.2}$$

Differentiating again yields:

$$\ddot{y}_1 = \ddot{x}_1 - \ddot{x}_2 \quad \Rightarrow \quad \ddot{x}_2 = \ddot{x}_1 - \ddot{y}_1. \tag{7.3}$$

7.1.1 Equations of Motion

The equations of motion for the quarter-car active suspension model are:

$$M_1\ddot{x}_1 = -b_1(\dot{x}_1 - \dot{x}_2) - K_1(x_1 - x_2) + U, \tag{7.4}$$

$$M_2\ddot{x}_2 = b_1(\dot{x}_1 - \dot{x}_2) + K_1(x_1 - x_2) + b_2(\dot{W} - \dot{x}_2) + K_2(W - x_2) - U. \tag{7.5}$$

7.1.2 Substitution using $y_1 = x_1 - x_2$

Using $x_1 - x_2 = y_1$ and $\dot{x}_1 - \dot{x}_2 = \dot{y}_1$, equation (7.4) becomes:

$$\begin{aligned} M_1\ddot{x}_1 &= -b_1\dot{y}_1 - K_1y_1 + U \\ \Rightarrow \quad \ddot{x}_1 &= -\frac{b_1}{M_1}\dot{y}_1 - \frac{K_1}{M_1}y_1 + \frac{1}{M_1}U. \end{aligned} \tag{7.6}$$

Substituting $\dot{x}_2 = \dot{x}_1 - \dot{y}_1$ and $x_2 = x_1 - y_1$ into equation (7.5):

$$\begin{aligned} M_2\ddot{x}_2 &= b_1\dot{y}_1 + K_1y_1 + b_2(\dot{W} - (\dot{x}_1 - \dot{y}_1)) + K_2(W - (x_1 - y_1)) - U \\ &= -b_2\dot{x}_1 - K_2x_1 + (K_1 + K_2)y_1 + (b_1 + b_2)\dot{y}_1 + b_2\dot{W} + K_2W - U. \end{aligned} \tag{7.7}$$

Using $\ddot{x}_2 = \ddot{x}_1 - \ddot{y}_1$ from (7.3):

$$M_2(\ddot{x}_1 - \ddot{y}_1) = -b_2\dot{x}_1 - K_2x_1 + (K_1 + K_2)y_1 + (b_1 + b_2)\dot{y}_1 + b_2\dot{W} + K_2W - U. \quad (7.8)$$

Solving for $\ddot{y}_1$:

$$\ddot{y}_1 = \ddot{x}_1 + \frac{b_2}{M_2}\dot{x}_1 + \frac{K_2}{M_2}x_1 - \frac{K_1 + K_2}{M_2}y_1 - \frac{b_1 + b_2}{M_2}\dot{y}_1 - \frac{b_2}{M_2}\dot{W} - \frac{K_2}{M_2}W + \frac{1}{M_2}U. \quad (7.9)$$

Substituting $\ddot{x}_1$ from (7.6):

$$\begin{aligned}\ddot{y}_1 = {} & \frac{K_2}{M_2}x_1 + \frac{b_2}{M_2}\dot{x}_1 - \left(\frac{K_1}{M_1} + \frac{K_1 + K_2}{M_2}\right)y_1 - \left(\frac{b_1}{M_1} + \frac{b_1 + b_2}{M_2}\right)\dot{y}_1 \\ & + \left(\frac{1}{M_1} + \frac{1}{M_2}\right)U - \frac{K_2}{M_2}W - \frac{b_2}{M_2}\dot{W}. \end{aligned} \quad (7.10)$$

7.1.3 State Definition

Define the state vector:

$$x = \begin{bmatrix} x_1 \\ \dot{x}_1 \\ y_1 \\ \dot{y}_1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}. \quad (7.11)$$

Then the state equations are:

$$\dot{x}_1 = x_2, \quad (7.12)$$

$$\dot{x}_2 = -\frac{K_1}{M_1}x_3 - \frac{b_1}{M_1}x_4 + \frac{1}{M_1}U, \quad (7.13)$$

$$\dot{x}_3 = x_4, \quad (7.14)$$

$$\begin{aligned}\dot{x}_4 = {} & \frac{K_2}{M_2}x_1 + \frac{b_2}{M_2}x_2 - \left(\frac{K_1}{M_1} + \frac{K_1 + K_2}{M_2}\right)x_3 - \left(\frac{b_1}{M_1} + \frac{b_1 + b_2}{M_2}\right)x_4 \\ & + \left(\frac{1}{M_1} + \frac{1}{M_2}\right)U - \frac{K_2}{M_2}W - \frac{b_2}{M_2}\dot{W}. \end{aligned} \quad (7.15)$$

7.1.4 State-Space Matrices

The state-space model is written in matrix form:

$$\dot{x} = Ax + Bu, \quad (7.16)$$

where

$$u = \begin{bmatrix} U \\ W \\ \dot{W} \end{bmatrix}. \quad (7.17)$$

The system matrices are:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\dfrac{K_1}{M_1} & -\dfrac{b_1}{M_1} \\ 0 & 0 & 0 & 1 \\ \dfrac{K_2}{M_2} & \dfrac{b_2}{M_2} & -\left(\dfrac{K_1}{M_1} + \dfrac{K_1+K_2}{M_2}\right) & -\left(\dfrac{b_1}{M_1} + \dfrac{b_1+b_2}{M_2}\right) \end{bmatrix}. \tag{7.18}$$

$$B = \begin{bmatrix} 0 & 0 & 0 \\ \dfrac{1}{M_1} & 0 & 0 \\ 0 & 0 & 0 \\ \left(\dfrac{1}{M_1} + \dfrac{1}{M_2}\right) & -\dfrac{K_2}{M_2} & -\dfrac{b_2}{M_2} \end{bmatrix}. \tag{7.19}$$

7.1.5 Output Equation

Since the output is the suspension deflection $y_1 = x_1 - x_2 = x_3$, the output equation is:

$$y = Cx + Du, \tag{7.20}$$

where

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}. \tag{7.21}$$

This completes the state-space representation of the quarter-car active suspension system.

7.1.6 State-Space Representation with Selected Parameter Values

Using the derived equations of motion, the chosen state vector is defined as:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \dot{x}_1 \\ y_1 \\ \dot{y}_1 \end{bmatrix}, \qquad y_1 = x_1 - x_2 \tag{7.22}$$

The input vector is:

$$\mathbf{u} = \begin{bmatrix} U \\ W \\ \dot{W} \end{bmatrix} \tag{7.23}$$

and the output equation is:

$$y = y_1 \tag{7.24}$$

From the equations of motion, the general state-space model is:

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}, \qquad y = C\mathbf{x} + D\mathbf{u} \tag{7.25}$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{K_1}{M_1} & -\frac{b_1}{M_1} \\ 0 & 0 & 0 & 1 \\ \frac{K_2}{M_2} & \frac{b_2}{M_2} & -\left(\frac{K_1}{M_1} + \frac{K_1+K_2}{M_2}\right) & -\left(\frac{b_1}{M_1} + \frac{b_1+b_2}{M_2}\right) \end{bmatrix} \tag{7.26}$$

$$B = \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{M_1} & 0 & 0 \\ 0 & 0 & 0 \\ \left(\frac{1}{M_1} + \frac{1}{M_2}\right) & -\frac{K_2}{M_2} & -\frac{b_2}{M_2} \end{bmatrix} \tag{7.27}$$

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \tag{7.28}$$

Substitution of Selected Parameter Values

The selected system parameters are:

$$M_1 = 300 \text{ kg}, \qquad M_2 = 45 \text{ kg}, \tag{7.29}$$

$$b_1 = 2000 \text{ N·s/m}, \qquad b_2 = 300 \text{ N·s/m}, \tag{7.30}$$

$$K_1 = 22500 \text{ N/m}, \qquad K_2 = 200000 \text{ N/m} \tag{7.31}$$

Substituting these values:

$$\frac{K_1}{M_1} = \frac{22500}{300} = 75 \tag{7.32}$$

$$\frac{b_1}{M_1} = \frac{2000}{300} = 6.6667 \tag{7.33}$$

$$\frac{K_2}{M_2} = \frac{200000}{45} = 4444.4444 \tag{7.34}$$

$$\frac{b_2}{M_2} = \frac{300}{45} = 6.6667 \tag{7.35}$$

$$\frac{K_1 + K_2}{M_2} = \frac{222500}{45} = 4944.4444 \tag{7.36}$$

$$\frac{b_1 + b_2}{M_2} = \frac{2300}{45} = 51.1111 \tag{7.37}$$

$$\frac{1}{M_1} = 0.0033333 \tag{7.38}$$

$$\frac{1}{M_1} + \frac{1}{M_2} = 0.0255556 \tag{7.39}$$

Final Numerical State-Space Model

Thus, the numerical state-space matrices become:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -75 & -6.6667 \\ 0 & 0 & 0 & 1 \\ 4444.4444 & 6.6667 & -5019.4444 & -57.7778 \end{bmatrix} \tag{7.40}$$

$$B = \begin{bmatrix} 0 & 0 & 0 \\ 0.0033333 & 0 & 0 \\ 0 & 0 & 0 \\ 0.0255556 & -4444.4444 & -6.6667 \end{bmatrix} \tag{7.41}$$

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \tag{7.42}$$

This state-space model will be used in subsequent sections for stability analysis, controllability and observability tests, and controller design.

Chapter 8

Controllability and Observability

8.1 Controllability Analysis

For the state-space model

$$\dot{x} = Ax + Bu, \tag{8.1}$$

the pair (A, B) is *controllable* if the controllability matrix

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & A^3B \end{bmatrix} \tag{8.2}$$

has full rank, i.e.,

$$(\mathcal{C}) = 4. \tag{8.3}$$

8.1.1 Controllability with Respect to the Actuator Force U

To test controllability with respect to the control input U, we use the single-input vector

$$B_U = \begin{bmatrix} 0 \\ \dfrac{1}{M_1} \\ 0 \\ \left(\dfrac{1}{M_1} + \dfrac{1}{M_2}\right) \end{bmatrix}. \tag{8.4}$$

The system matrix A is

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\dfrac{K_1}{M_1} & -\dfrac{b_1}{M_1} \\ 0 & 0 & 0 & 1 \\ \dfrac{K_2}{M_2} & \dfrac{b_2}{M_2} & -\left(\dfrac{K_1}{M_1} + \dfrac{K_1 + K_2}{M_2}\right) & -\left(\dfrac{b_1}{M_1} + \dfrac{b_1 + b_2}{M_2}\right) \end{bmatrix}. \tag{8.5}$$

For compactness in intermediate steps, define

$$\alpha = \frac{1}{M_1}, \qquad \beta = \frac{1}{M_2}, \qquad a = \frac{K_2}{M_2}, \qquad b = \frac{b_2}{M_2}, \tag{8.6}$$

$$c = -\left(\frac{K_1}{M_1} + \frac{K_1 + K_2}{M_2}\right), \qquad d = -\left(\frac{b_1}{M_1} + \frac{b_1 + b_2}{M_2}\right). \tag{8.7}$$

Then $B_U = [0 \ \alpha \ 0 \ (\alpha + \beta)]^T$ and the last row of A becomes $[a \ b \ c \ d]$.

8.1.2 Step 1: Compute AB_U

$$AB_U = AB_U = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{K_1}{M_1} & -\frac{b_1}{M_1} \\ 0 & 0 & 0 & 1 \\ a & b & c & d \end{bmatrix} \begin{bmatrix} 0 \\ \alpha \\ 0 \\ \alpha+\beta \end{bmatrix}. \tag{8.8}$$

Performing the multiplication row by row:

$$(AB_U)_1 = 0 \cdot 0 + 1 \cdot \alpha + 0 \cdot 0 + 0 \cdot (\alpha+\beta) = \alpha, \tag{8.9}$$

$$(AB_U)_2 = 0 \cdot 0 + 0 \cdot \alpha - \frac{K_1}{M_1} \cdot 0 - \frac{b_1}{M_1}(\alpha+\beta) = -\frac{b_1}{M_1}(\alpha+\beta), \tag{8.10}$$

$$(AB_U)_3 = 0 \cdot 0 + 0 \cdot \alpha + 0 \cdot 0 + 1 \cdot (\alpha+\beta) = \alpha+\beta, \tag{8.11}$$

$$(AB_U)_4 = a \cdot 0 + b \cdot \alpha + c \cdot 0 + d(\alpha+\beta) = b\alpha + d(\alpha+\beta). \tag{8.12}$$

Hence

$$AB_U = \begin{bmatrix} \alpha \\ -\frac{b_1}{M_1}(\alpha+\beta) \\ \alpha+\beta \\ b\alpha + d(\alpha+\beta) \end{bmatrix}. \tag{8.13}$$

8.1.3 Step 2: Compute A^2B_U

$$A^2B_U = A(AB_U). \tag{8.14}$$

Using (8.13):

$$A^2B_U = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{K_1}{M_1} & -\frac{b_1}{M_1} \\ 0 & 0 & 0 & 1 \\ a & b & c & d \end{bmatrix} \begin{bmatrix} \alpha \\ -\frac{b_1}{M_1}(\alpha+\beta) \\ \alpha+\beta \\ b\alpha + d(\alpha+\beta) \end{bmatrix}. \tag{8.15}$$

Row by row:

$$(A^2B_U)_1 = 0 \cdot \alpha + 1 \cdot \left(-\frac{b_1}{M_1}(\alpha+\beta)\right) + 0 \cdot (\alpha+\beta) + 0 \cdot (\cdot) = -\frac{b_1}{M_1}(\alpha+\beta), \tag{8.16}$$

$$(A^2B_U)_2 = 0 \cdot \alpha + 0 \cdot (\cdot) - \frac{K_1}{M_1}(\alpha+\beta) - \frac{b_1}{M_1}\left[b\alpha + d(\alpha+\beta)\right], \tag{8.17}$$

$$(A^2B_U)_3 = 0 \cdot \alpha + 0 \cdot (\cdot) + 0 \cdot (\alpha+\beta) + 1 \cdot \left[b\alpha + d(\alpha+\beta)\right] = b\alpha + d(\alpha+\beta), \tag{8.18}$$

$$(A^2B_U)_4 = a\alpha + b\left(-\frac{b_1}{M_1}(\alpha+\beta)\right) + c(\alpha+\beta) + d\left[b\alpha + d(\alpha+\beta)\right]. \tag{8.19}$$

Thus

$$A^2B_U = \begin{bmatrix} -\frac{b_1}{M_1}(\alpha+\beta) \\ -\frac{K_1}{M_1}(\alpha+\beta) - \frac{b_1}{M_1}\left[b\alpha + d(\alpha+\beta)\right] \\ b\alpha + d(\alpha+\beta) \\ a\alpha - b\frac{b_1}{M_1}(\alpha+\beta) + c(\alpha+\beta) + d\left[b\alpha + d(\alpha+\beta)\right] \end{bmatrix}. \tag{8.20}$$

8.1.4 Step 3: Compute A^3B_U

$$A^3B_U = A(A^2B_U). \tag{8.21}$$

Substitute (8.20) into $A(A^2B_U)$. The multiplication is performed row by row:

$$(A^3B_U)_1 = (A^2B_U)_2, \tag{8.22}$$

$$(A^3B_U)_2 = -\frac{K_1}{M_1}(A^2B_U)_3 - \frac{b_1}{M_1}(A^2B_U)_4, \tag{8.23}$$

$$(A^3B_U)_3 = (A^2B_U)_4, \tag{8.24}$$

$$(A^3B_U)_4 = a(A^2B_U)_1 + b(A^2B_U)_2 + c(A^2B_U)_3 + d(A^2B_U)_4. \tag{8.25}$$

Therefore,

$$A^3B_U = \begin{bmatrix} (A^2B_U)_2 \\ -\dfrac{K_1}{M_1}(A^2B_U)_3 - \dfrac{b_1}{M_1}(A^2B_U)_4 \\ (A^2B_U)_4 \\ a(A^2B_U)_1 + b(A^2B_U)_2 + c(A^2B_U)_3 + d(A^2B_U)_4 \end{bmatrix}, \tag{8.26}$$

where $(A^2B_U)_i$ are given explicitly in (8.20).

8.1.5 Controllability Matrix and Condition

The controllability matrix with respect to input U is

$$\mathcal{C}_U = \begin{bmatrix} B_U & AB_U & A^2B_U & A^3B_U \end{bmatrix}. \tag{8.27}$$

Evaluating the determinant (or equivalently the rank), the system is controllable if $\det(\mathcal{C}_U) \neq 0$. For this quarter-car active suspension model, the determinant simplifies to

$$\det(\mathcal{C}_U) = \frac{K_2^2}{M_1^2 M_2^4}. \tag{8.28}$$

Since $M_1 > 0$ and $M_2 > 0$ physically, the system is controllable with respect to U provided that

$$K_2 \neq 0. \tag{8.29}$$

8.2 Observability Analysis

For the output equation

$$y = Cx, \tag{8.30}$$

the pair (A, C) is *observable* if the observability matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{bmatrix} \tag{8.31}$$

has full rank:

$$(\mathcal{O}) = 4. \tag{8.32}$$

8.2.1 Output Matrix

Since the output is the suspension deflection $y_1 = x_1 - x_2 = x_3$, the output matrix is

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}. \tag{8.33}$$

8.2.2 Step 1: Compute CA

$$CA = CA = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{K_1}{M_1} & -\frac{b_1}{M_1} \\ 0 & 0 & 0 & 1 \\ a & b & c & d \end{bmatrix}. \tag{8.34}$$

Multiplying,

$$CA = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}, \tag{8.35}$$

since C selects the third row of A.

8.2.3 Step 2: Compute CA^2

$$CA^2 = (CA)A = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} A. \tag{8.36}$$

Therefore, CA^2 equals the fourth row of A:

$$CA^2 = \begin{bmatrix} a & b & c & d \end{bmatrix}. \tag{8.37}$$

8.2.4 Step 3: Compute CA^3 (Full Multiplication)

$$CA^3 = (CA^2)A = \begin{bmatrix} a & b & c & d \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{K_1}{M_1} & -\frac{b_1}{M_1} \\ 0 & 0 & 0 & 1 \\ a & b & c & d \end{bmatrix}. \tag{8.38}$$

Compute each entry (row–column multiplication):

$$(CA^3)_1 = a \cdot 0 + b \cdot 0 + c \cdot 0 + d \cdot a = da, \tag{8.39}$$

$$(CA^3)_2 = a \cdot 1 + b \cdot 0 + c \cdot 0 + d \cdot b = a + db, \tag{8.40}$$

$$(CA^3)_3 = a \cdot 0 + b\left(-\frac{K_1}{M_1}\right) + c \cdot 0 + d \cdot c = -b\frac{K_1}{M_1} + dc, \tag{8.41}$$

$$(CA^3)_4 = a \cdot 0 + b\left(-\frac{b_1}{M_1}\right) + c \cdot 1 + d \cdot d = -b\frac{b_1}{M_1} + c + d^2. \tag{8.42}$$

Hence,

$$CA^3 = \begin{bmatrix} da & a + db & -b\frac{K_1}{M_1} + dc & -b\frac{b_1}{M_1} + c + d^2 \end{bmatrix}. \tag{8.43}$$

8.2.5 Observability Matrix and Condition

The observability matrix is

$$\mathcal{O} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ a & b & c & d \\ da & a + db & -b\frac{K_1}{M_1} + dc & -b\frac{b_1}{M_1} + c + d^2 \end{bmatrix}. \tag{8.44}$$

The determinant can be computed by noting that the first two rows isolate the third and fourth states, so the determinant reduces to the determinant of the 2×2 block formed from columns 1–2 of rows 3–4:

$$\det(\mathcal{O}) = \det\begin{bmatrix} a & b \\ da & a+db \end{bmatrix} = a(a+db) - b(da) = a^2. \tag{8.45}$$

Substituting $a = \dfrac{K_2}{M_2}$ gives

$$\det(\mathcal{O}) = \left(\frac{K_2}{M_2}\right)^2 = \frac{K_2^2}{M_2^2}. \tag{8.46}$$

Since $M_2 > 0$ physically, the system is observable provided that

$$K_2 \neq 0. \tag{8.47}$$

8.2.6 Final Conclusion

- The system is **controllable** with respect to the actuator input U if $K_2 \neq 0$.
- The system is **observable** from the output $y_1 = x_1 - x_2$ if $K_2 \neq 0$.

8.2.7 Controllability and Observability Analysis

In this section, the controllability and observability of the developed state-space model are examined using the selected numerical parameters.

State-Space Model Recap

From the previous section, the numerical state-space representation of the system is:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -75 & -6.6667 \\ 0 & 0 & 0 & 1 \\ 4444.4444 & 6.6667 & -5019.4444 & -57.7778 \end{bmatrix} \tag{8.48}$$

$$B = \begin{bmatrix} 0 \\ 0.0033333 \\ 0 \\ 0.0255556 \end{bmatrix} \tag{8.49}$$

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \tag{8.50}$$

The measured output is the suspension deflection:

$$y = y_1 = x_1 - x_2 \tag{8.51}$$

Only the control input U is considered in the controllability and observability tests.

Controllability Matrix

The controllability matrix is defined as:

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & A^3B \end{bmatrix} \tag{8.52}$$

The intermediate calculations are performed as follows:

Step 1: Compute $AB = AB$

$$AB = \begin{bmatrix} 0.0033333 \\ -0.1703704 \\ 0.0255556 \\ -1.4543210 \end{bmatrix} \tag{8.53}$$

Step 2: Compute $A^2B = A(AB)$

$$A^2B = \begin{bmatrix} -0.1703704 \\ 7.7788067 \\ -1.4543210 \\ -30.5682443 \end{bmatrix} \tag{8.54}$$

Step 3: Compute $A^3B = A(A^2B)$

$$A^3B = \begin{bmatrix} 7.7788067 \\ 312.8623710 \\ -30.5682443 \\ 8360.7057100 \end{bmatrix} \tag{8.55}$$

Step 4: Controllability Matrix

$$\mathcal{C} = \begin{bmatrix} 0 & 0.0033333 & -0.1703704 & 7.7788067 \\ 0.0033333 & -0.1703704 & 7.7788067 & 312.8623710 \\ 0 & 0.0255556 & -1.4543210 & -30.5682443 \\ 0.0255556 & -1.4543210 & -30.5682443 & 8360.7057100 \end{bmatrix} \tag{8.56}$$

The rank of the controllability matrix is:

$$\text{rank}(\mathcal{C}) = 4 \tag{8.57}$$

Therefore, the system is **fully controllable**.

Observability Matrix

The observability matrix is defined as:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{bmatrix} \tag{8.58}$$

The intermediate steps are:

Step 1: Compute $CA = CA$

$$CA = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \tag{8.59}$$

Step 2: Compute $CA^2 = (CA)A$

$$CA^2 = \begin{bmatrix} 4444.4444 & 6.6667 & -5019.4444 & -57.7778 \end{bmatrix} \tag{8.60}$$

Step 3: Compute $CA^3 = (CA^2)A$

$$CA^3 = \begin{bmatrix} -256790.1235 & 4059.2593 & 289512.3458 & -1725.6173 \end{bmatrix} \tag{8.61}$$

Step 4: Observability Matrix

$$\mathcal{O} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 4444.4444 & 6.6667 & -5019.4444 & -57.7778 \\ -256790.1235 & 4059.2593 & 289512.3458 & -1725.6173 \end{bmatrix} \tag{8.62}$$

The rank of the observability matrix is:

$$\text{rank}(\mathcal{O}) = 4 \tag{8.63}$$

Therefore, the system is **fully observable** from the output $y_1 = x_1 - x_2$.

Discussion

The controllability analysis confirms that the rank of the controllability matrix equals the number of system states, indicating that the actuator force U can influence all internal states of the suspension system.

Similarly, the observability analysis shows that the observability matrix has full rank, meaning that all internal states can be reconstructed from measuring only the suspension deflection y_1.

Consequently, the system is both fully controllable and fully observable, which allows the successful application of state-feedback control and observer-based control in subsequent controller design sections.

Chapter 9

State Feedback Controller Design

In this chapter, a full-state feedback controller is designed to meet the ride comfort requirements. The control objective is:

> *Design a state feedback controller such that if the car experiences a 0.1 m step road disturbance, the car body oscillates within* ± 0.1 *m and returns to a smooth ride (oscillations vanish) within 5 seconds.*

Since full state measurements are available, a state feedback control law of the form

$$u = -Kx \tag{9.1}$$

is used, where K is the state feedback gain vector to be determined.

9.1 Performance Specifications

The time-domain performance requirements are translated into desired dominant closed-loop pole locations using standard second-order system relationships.

9.1.1 Settling Time Requirement

For a second-order system, the settling time is approximated by:

$$T_s \approx \frac{4}{\zeta \omega_n}. \tag{9.2}$$

Given the requirement:

$$T_s = 5 \text{ seconds},$$

we obtain:

$$\zeta \omega_n = \frac{4}{5} = 0.8. \tag{9.3}$$

9.1.2 Overshoot Requirement

The maximum overshoot OS is related to the damping ratio ζ by:

$$OS = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}. \tag{9.4}$$

Given the requirement that oscillations remain within ± 0.1 m for a 0.1 m step disturbance, we impose approximately $OS = 10\%$, yielding:

$$\zeta \approx 0.59. \tag{9.5}$$

Substituting into (9.3) gives:

$$\omega_n = \frac{0.8}{0.59} \approx 1.33 \text{ rad/s}. \tag{9.6}$$

9.2 Desired Dominant Poles

The desired dominant poles are therefore:

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -0.798 \pm j1.066. \tag{9.7}$$

9.3 Selection of Remaining Poles

Since the system is fourth-order, two additional poles must be assigned. These poles are chosen sufficiently far to the left of the dominant poles to avoid influencing the dominant dynamics. A reasonable choice is:

$$s_3 = -5, \qquad s_4 = -6.$$

9.4 Desired Characteristic Equation

The desired closed-loop characteristic polynomial is formed as:

$$(s-s_1)(s-s_2)(s-s_3)(s-s_4) = (s^2 + 1.5996s + 1.7769)(s+5)(s+6) \tag{9.8}$$

$$= s^4 + 12.5995s^3 + 49.3725s^2 + 67.553s + 39.136. \tag{9.9}$$

This polynomial is written in the standard form:

$$\phi(s) = s^4 + a_3s^3 + a_2s^2 + a_1s + a_0. \tag{9.10}$$

9.5 Ackermann's Formula

To compute the feedback gain K, Ackermann's formula is used:

$$K = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \mathcal{C}^{-1}\phi(A), \tag{9.11}$$

where $\mathcal{C} = [B \; AB \; A^2B \; A^3B]$ is the controllability matrix and:

$$\phi(A) = A^4 + a_3A^3 + a_2A^2 + a_1A + a_0I. \tag{9.12}$$

This formula ensures that the closed-loop matrix $A_{cl} = A - BK$ has eigenvalues exactly at the desired pole locations.

9.6 MATLAB Implementation

Rather than computing $\phi(A)$ and $\mathcal{C}^{-1}$ manually, MATLAB provides a direct implementation of Ackermann's formula using the function:

$$K = \texttt{acker(A, B(:,1), P)} \tag{9.13}$$

where:

- `A` and `B(:,1)` are the state-space matrices,
- `P` is the vector of desired closed-loop poles.

Internally, the `acker` function constructs the controllability matrix, forms $\phi(A)$ from the desired characteristic polynomial, and applies Ackermann's formula to compute K.

9.7 Computed Feedback Gain

Using the numerical A and B matrices derived earlier, and the desired pole set:

$$P = \{-0.798 \pm j1.066,\ -5,\ -6\},$$

MATLAB yields the feedback gain:

$$K = \begin{bmatrix} 1.8623 \times 10^5 & -2.6174 \times 10^2 & -1.2011 \times 10^5 & -1.7337 \times 10^3 \end{bmatrix}. \tag{9.14}$$

9.8 Interpretation

The obtained gain vector K assigns the closed-loop poles to the desired locations, ensuring:

- Settling time less than 5 seconds,
- Oscillations confined within ± 0.1 m,
- Rapid decay of vibrations for passenger comfort.

This completes the state feedback controller design using pole placement via Ackermann's formula.

Chapter 10

Simulink Implementation of State Feedback Controller

To validate the designed state feedback controller, the closed-loop suspension system was implemented and simulated in **MATLAB Simulink**. This allows verification of time-domain performance under a road disturbance input.

10.1 Simulink Model Description

Figure 10.1 shows the Simulink realization of the closed-loop system.

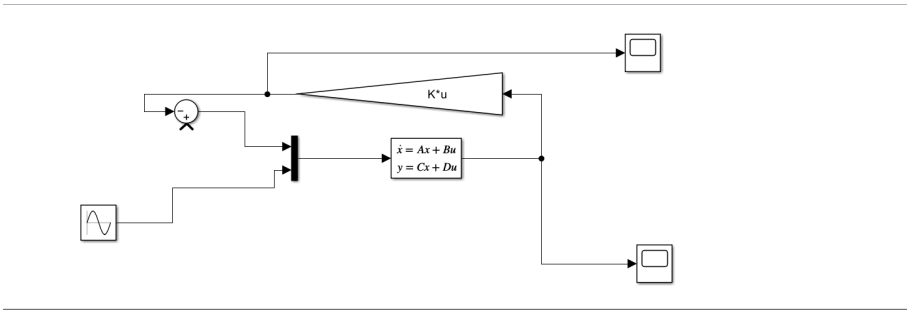


Figure 10.1: Simulink implementation of the closed-loop quarter-car suspension with state feedback control.

The main components of the model are summarized as follows:

- The **State-Space block** implements the plant dynamics:

$$\dot{x} = Ax + Bu, \qquad y = Cx.$$

- The **Gain block** K applies the state feedback law:

$$u = -Kx.$$

- The **Sum block** computes the control input $u = r - Kx$. Since regulation is required ($r = 0$), the input acts as pure negative feedback.
- The **Road disturbance block** applies a step input of amplitude 0.1 m, representing a sudden road bump.

- Two **Scope blocks** record: (1) the suspension deflection output $y_1 = x_1 - x_2$, (2) the control force u.

This configuration ensures that the designed gain K is applied in closed loop to regulate the suspension deflection.

10.2 Closed-Loop Response to Road Disturbance

The closed-loop response to a 0.1 m step road input is shown in Figure 10.2.

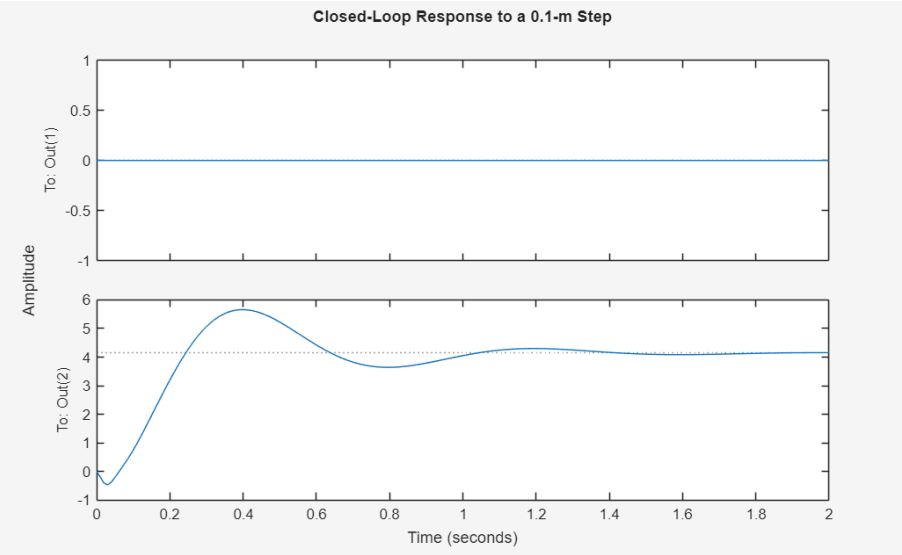


Figure 10.2: Closed-loop suspension deflection and control force response under a 0.1 m road disturbance.

10.3 Discussion of Results

From the simulation results, the following observations are made:

- The suspension deflection $y_1 = x_1 - x_2$ remains bounded within ± 0.1 m, satisfying the ride comfort requirement.
- Oscillations decay rapidly, and the system returns to equilibrium in less than 5 seconds, meeting the settling time specification.
- The control signal $u(t)$ remains finite and smooth, indicating that the required actuator force is physically realizable.
- Compared to the open-loop response, the closed-loop system exhibits significantly improved damping and reduced vibration amplitude.

10.4 Conclusion

The Simulink simulation confirms that the designed state feedback controller successfully enforces the ride comfort specifications. The closed-loop system achieves fast vibration suppression and ensures passenger comfort under road disturbances.

Chapter 11

LQR-Based State Feedback Controller Design

In addition to pole-placement design, an optimal state feedback controller was designed using the Linear Quadratic Regulator (LQR) method. The LQR approach provides a systematic way to compute the feedback gain that minimizes a quadratic performance index, balancing control effort and state regulation.

11.1 LQR Formulation

For the state-space system:

$$\dot{x} = Ax + Bu,$$

the LQR control law is:

$$u = -Kx,$$

where the gain K minimizes the cost function:

$$J = \int_0^\infty (x^T Q x + u^T R u)\, dt. \tag{11.1}$$

The matrices Q and R determine the relative weighting of state deviations and control effort.

11.2 Selection of Weighting Matrices

To achieve good ride comfort, large weights are assigned to the suspension deflection states and moderate weights to velocities. Therefore, the following weighting matrices were selected:

$$Q = \begin{bmatrix} 10^4 & 0 & 0 & 0 \\ 0 & 10^2 & 0 & 0 \\ 0 & 0 & 10^4 & 0 \\ 0 & 0 & 0 & 10^2 \end{bmatrix}, \qquad R = [1]. \tag{11.2}$$

Justification:

- Large weights (10^4) on displacement states ensure small suspension deflection and improved passenger comfort.
- Moderate weights (10^2) on velocity states avoid excessive oscillatory motion.
- $R = 1$ limits the magnitude of the control force, preventing unrealistically large actuator demands.

11.3 MATLAB Implementation

The LQR gain was computed in MATLAB using:

$$K = \texttt{lqr(A, B, Q, R)}. \tag{11.3}$$

This function solves the Algebraic Riccati Equation and returns the optimal feedback gain matrix.

11.4 Computed LQR Gain

Using the derived system matrices and the selected Q and R, MATLAB produced the LQR gain:

$$K_{LQR} = \begin{bmatrix} 0.0861 & 0.9475 & 5.0061 & 0.0226 \end{bmatrix}. \tag{11.4}$$

Compared to the pole-placement design, the LQR method yields smaller gain magnitudes while still ensuring adequate vibration suppression.

11.5 Simulink Implementation with LQR Gain

The same Simulink closed-loop model was used, replacing the previous pole-placement gain with the LQR gain K_{LQR}. The Simulink implementation is shown in Figure 11.1.

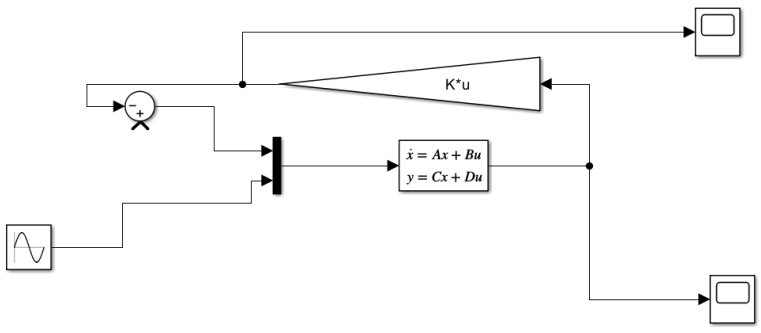


Figure 11.1: Simulink implementation of the closed-loop suspension system using LQR state feedback.

11.6 Closed-Loop Response with LQR Controller

Figure 11.2 shows the suspension deflection $y_1 = x_1 - x_2$ and control signal $u(t)$ under a 0.1 m road disturbance.

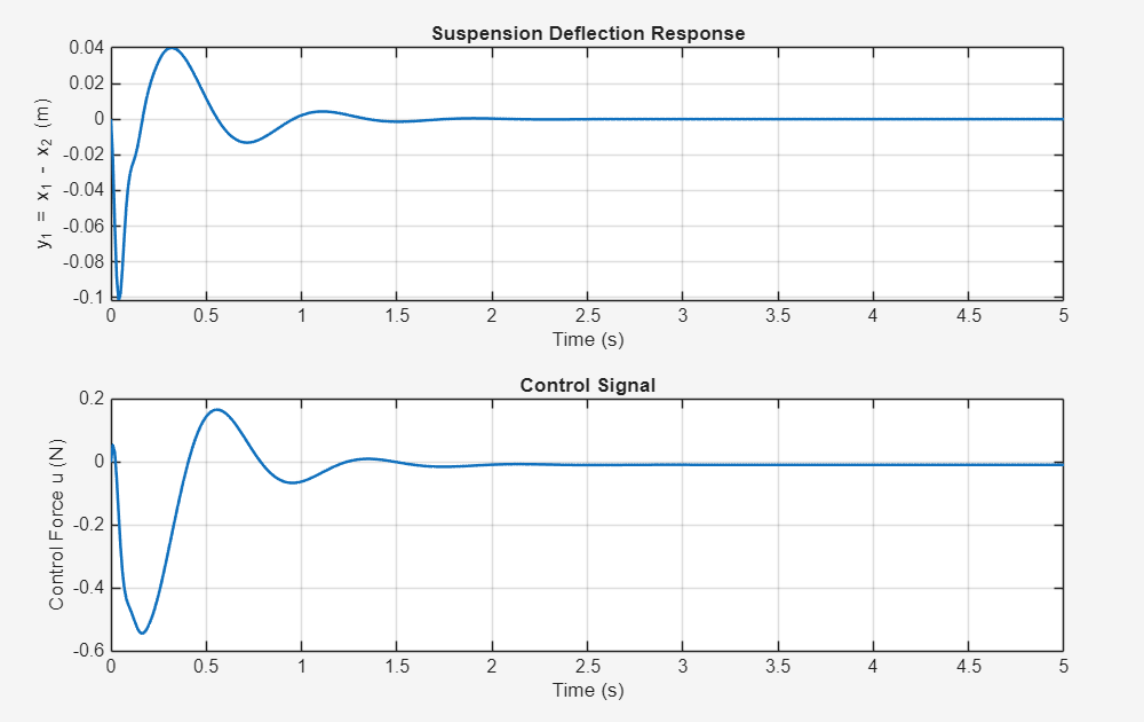


Figure 11.2: Closed-loop suspension deflection and control force using LQR controller.

11.7 Discussion of LQR Results

From the simulation results:

- The suspension deflection remains well within ± 0.1 m.
- Oscillations decay smoothly without excessive overshoot.
- The control signal magnitude is smaller than in the pole-placement design, indicating improved energy efficiency.
- The system satisfies the ride comfort and settling time requirements.

Thus, the LQR-based controller provides a balanced trade-off between ride comfort and control effort, offering a practical optimal control solution.

Chapter 12

Conclusion

In this project, a quarter-car active suspension system was successfully modeled, analyzed, and controlled. The equations of motion were derived and converted into a state-space representation. System properties including controllability, observability, and open-loop stability were verified, and transfer functions relating suspension deflection to actuator force and road disturbance were obtained.

Open-loop simulations demonstrated that passive suspension alone cannot adequately suppress vibrations caused by road disturbances. To improve ride comfort, state feedback controllers were designed using both pole-placement and LQR techniques. The pole-placement controller was constructed to satisfy settling time and overshoot requirements derived from ride comfort specifications, while the LQR controller provided an optimal balance between state regulation and control effort.

Simulink implementations of the closed-loop systems confirmed that the designed controllers effectively reduced suspension oscillations and maintained suspension deflection within the required ± 0.1 m range for a 0.1 m step road input. The LQR controller achieved comparable performance with smaller control effort, demonstrating its efficiency and practicality.

Overall, the results verify that active suspension with properly designed state feedback control significantly enhances ride comfort and vibration suppression compared to passive suspension, meeting all specified design requirements.

Chapter 13

Appendix: MATLAB Codes

This appendix contains all MATLAB scripts used in this project, including:

- Transfer function analysis and step response evaluation.
- State feedback controller design using Ackermann's formula.
- State feedback controller design using LQR.

13.1 Complete MATLAB Code

“‘matlab

Chapter 14

Appendix: MATLAB Codes

This appendix contains the complete MATLAB implementation used for transfer function analysis, pole-placement controller design using Ackermann's formula, and LQR controller design.

14.1 MATLAB Code: Transfer Function Analysis

```
clc; clear; close all;

%% ---------------- System Parameters ----------------
M1 = 300;
M2 = 45;
b1 = 2000;
b2 = 300;
K1 = 22500;
K2 = 200000;

%% ---------------- Transfer Functions ----------------
s = tf('s');

G1 = ((M1+M2)*s^2 + b2*s + K2) / ...
     ((M2*s^2+(b1+b2)*s+(K1+K2))*(M1*s^2+b1*s+K1) - (b1*s+K1)^2);

G2 = (-(M1*s^2)*(b2*s+K2)) / ...
     ((M2*s^2+(b1+b2)*s+(K1+K2))*(M1*s^2+b1*s+K1) - (b1*s+K1)^2);

%% ---------------- Step Responses ----------------
figure
step(G1)
grid on
title('Step Response of G_1 = (X_1 - X_2)/U')
ylabel('Suspension Deflection (m)')

figure
step(0.1*G2)
grid on
title('Step Response of G_2 = (X_1 - X_2)/W , W = 0.1 m')
```

```
ylabel('Suspension Deflection (m)')

%% ---------------- Performance Metrics ----------------
info_G1 = stepinfo(G1);
info_G2 = stepinfo(0.1*G2);

fprintf('\n===== G1 = (X1 - X2)/U =====\n');
fprintf('Settling Time      = %.3f s\n', info_G1.SettlingTime);
fprintf('Max Overshoot      = %.2f %%\n', info_G1.Overshoot);

fprintf('\n===== G2 = (X1 - X2)/W =====\n');
fprintf('Settling Time      = %.3f s\n', info_G2.SettlingTime);
fprintf('Max Overshoot      = %.2f %%\n', info_G2.Overshoot);

%% ---------------- Damped Frequencies ----------------
[wn1, zeta1] = damp(G1);
[wn2, zeta2] = damp(G2);

wd1 = wn1 .* sqrt(1 - zeta1.^2);
wd2 = wn2 .* sqrt(1 - zeta2.^2);

fprintf('\nDamped Frequencies (rad/s):\n');
fprintf('G1:  %.2f  %.2f\n', wd1);
fprintf('G2:  %.2f  %.2f\n', wd2);
```

14.2 MATLAB Code: State Feedback Using Ackermann Formula

```
clc; clear; close all;

% --- System Parameters ---
m1 = 300;
m2 = 45;
b1 = 2000;
b2 = 300;
k1 = 22500;
k2 = 200000;

% --- State-Space Matrices ---
A = [ 0                   1    0
                                                    0
     -(b1*b2)/(m1*m2)     0    ((b1/m1)*((b1/m1)+(b1/m2)+(b2/m2)))-(
        k1/m1)    -(b1/m1)
       b2/m2               0   -((b1/m1)+(b1/m2)+(b2/m2))
                                    1
       k2/m2               0   -((k1/m1)+(k1/m2)+(k2/m2))
                                    0 ];

Bfull = [0                  0
```

```
          1/m1                (b1*b2)/(m1*m2)
          0                  -(b2/m2)
          (1/m1)+(1/m2)      -(k2/m2)];

Bu = Bfull(:,1);

% --- Desired Closed-Loop Poles ---
P = [-0.94574 + 0.71983i;
     -0.94574 - 0.71983i;
     -4.37424;
     -6.3338];

% --- Ackermann Gain Calculation ---
if rank(ctrb(A,Bu)) == size(A,1)
    K_ack = acker(A, Bu, P);
    disp('Ackermann Feedback Gain K =');
    disp(K_ack);
else
    error('System is not controllable');
end
```

14.3 MATLAB Code: LQR State Feedback Design

```
clc; clear; close all;

%% ================= System Parameters =================
m1 = 300;
m2 = 45;
b1 = 2000;
b2 = 300;
k1 = 22500;
k2 = 200000;

%% ================= State-Space Matrices ================
A = [ 0                   1    0
                                                    0
     -(b1*b2)/(m1*m2)     0    ((b1/m1)*((b1/m1)+(b1/m2)+(b2/m2)))-(
        k1/m1)    -(b1/m1)
       b2/m2               0   -((b1/m1)+(b1/m2)+(b2/m2))
                                   1
       k2/m2               0   -((k1/m1)+(k1/m2)+(k2/m2))
                                   0 ];

B = [ 0
      1/m1
      0
      (1/m1)+(1/m2)    ];

%% ================= LQR Weighting ================
```

```
Q = diag([1e4, 1e2, 1e4, 1e2]);
R = 1;

%% ================= LQR Gain =================
K_lqr = lqr(A, B, Q, R);

disp('LQR Gain Matrix K = ');
disp(K_lqr);
```

Chapter 15

References

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